Arush Vyas

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Education

Boston University — Boston, MA
Bachelor of Science in Biomedical Engineering | ***GPA**: 3.83* — *Expected Grad: May 2027*

Technical Skills

Programming: Python, PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, OpenCV, Matplotlib, MLflow, AWS, Git, R, Linux
Medical Imaging: QuPath, ImageJ, 3D Slicer, OpenSlide, wsInfer, Biopython, Groovy, DICOM, MONAI, CTA Image Analysis
Laboratory: Tissue Culture, Microscopy, Spectroscopy, Gel Imaging, Micro Pipetting, qPCR, Sterile Cell Culture, Bioprinting

Relevant Experience

Health AI Engineering Intern — May 2026 – Present
Boston Scientific Corporation — *Boston, MA*

- Developed and optimized an internal 3D CTA segmentation pipeline for a vascular-therapies AI medical-imaging product, accurately distinguishing 10+ artery classes across scans and achieving a mean artery Dice score of 0.86 and clDice of 0.94
- Trained 10+ variations by optimizing vessel-specific loss functions, attention mechanisms, and fine-tuning of foundation models
- Co-invented patent-pending medical-imaging technology, translating clinical requirements into an AI-driven technical solution

Undergraduate Deep Learning Researcher — Sep. 2025 – Present
Tian Lab | Computational Imaging Systems — *Boston, MA*

- Building the largest annotated dataset of CTE and Alzheimer's brain tissue histopathology at BU, with 1,000+ images of manually annotated weakly supervised labels designed to train models that pathologists can use for neuropathological feature detection
- Designed a vessel detection model that works across staining protocols for Brightfield H&E tissues, resulting in 92% accuracy driven by implementing modified Reinhard stain normalization, multi-head loss functions, and hyperparameter tuning

Lab Assistant — Sep. 2025 – Present
Boston University's Bioengineering Technology & Entrepreneurship Center — *Boston, MA*

- Implement and maintain an AI RAG-based knowledge base encompassing 1,800+ BTEC documents across 12 subject areas
- Prepare BTEC resources for AI retrieval by cleaning, restructuring, and converting files into consistent, model-ready formats
- Analyze usage patterns and allocate resource inventory across Biosensors, Tissue Engineering, and Digital Medicine Suites

Projects

VascuPath | *QuPath, Python, PyTorch, Groovy, WsInfer, OpenCV, OpenSlide, MLflow* — Jan. 2026 – Present

- Automating what currently takes neuropathologists hours of manual analysis: an AI pipeline that segments large-scale whole slide images to detect and quantify vascular changes in CTE and Alzheimer's, using 2,500+ tissue patches from real patients
- Developing a multi-head model architecture with stain separation, vessel detection, and vessel patch segmentation, designed to integrate with a neuropathology foundation model for scalable disease pattern analysis across neurodegenerative conditions
- Extracting and statistically analyzing 10+ vascular features, including tortuosity, diameter variation, branching density, and spatial distribution, to identify which patterns differ significantly across CTE and Alzheimer's stages (control, mild, severe)
- Designing an attention-based MIL framework that lets the model show which regions of a tissue slide drive its predictions

SymptomAI | *Python, Flask, React, PyTorch, Pandas, Scikit-Learn, NumPy, Matplotlib, OpenAI* — Oct. 2025

- Developed a disease prediction system tackling the hardest part of clinical classification: distinguishing between 773 diseases where symptoms heavily overlap across conditions, predicting the top 3 most likely conditions from 328 symptom inputs
- Engineered a PyTorch-based Convolutional Neural Network (CNN) to capture hidden symptom interaction patterns utilizing GPU acceleration and early stopping to achieve multi-label disease prediction, resulting in 82% accuracy (F1: 0.83, precision: 0.90)

Publications & Awards

- Vyas, A., Bansal, K., et al. "Optimizing Non-Invasive Brain-Computer Interfaces for Motor Rehabilitation in Individuals with Spinal Cord Injury." STEM Students on the Rise, Rice University, Oct. 2025. https://dx.doi.org/10.2139/ssrn.5605290
- 2x Student Research Award Recipient, Boston University's Undergraduate Research Opportunities Program (UROP), 2025, 2026
- Harold C. Case Merit Scholarship Recipient for Academic Excellence, Boston University, 2026, one of 15 selected recipients
- 3rd Place in Early-Stage Lung Cancer Phenotyping track, MedAI Hackathon, Boston University, April 2026
- 1st Place, Hackathon for Public Health Equity, Boston University's School of Public Health, April 2025